

SAVITRIBAI PHULE PUNE UNIVERSITY

Programming and Problem Solving (110005)

2019 Pattern | Semester I/II

Total Marks: 70 | Time: 2½ Hours

Instructions: Q1/Q2, Q3/Q4, Q5/Q6, Q7/Q8. Neat diagrams must be drawn wherever necessary.

Unit III --- Functions and Modules (18 Marks)

Q1) Attempt the following: [15]

- a) Define a **function**. Explain the need for functions in Python. [6]
- b) Differentiate between **local** and **global variables** with examples. [4]
- c) Write a program using a function to find the largest of 3 numbers by passing numbers as arguments. [5]

OR Q2) Attempt the following: [15]

- a) Explain the following types of function arguments with examples: i) **Default arguments** ii) **Variable length arguments** [6]
- b) Write a program to find the square of a number using **lambda function**. [5]
- c) What are **modules** and **packages** in Python? Explain a user-defined module with example. [4]

Unit IV --- Strings (17 Marks)

Q3) Attempt the following: [15]

- a) Explain different ways of formatting a **string** in Python with suitable examples. [6]
- b) Explain different ways of **iterating a string**. [4]
- c) What is the output of the following for string `S = "Learning Python is fun"`: i) `print(S[:10])` ii) `print(S[::-1])` iii) `print("Is" not in S)` iv) `print(S)` v) `print(S[0:10])` [5][4]

OR Q4) Attempt the following: [15]

- a) Explain the following string methods with examples: i) `isspace()` ii) `join()` iii) `lower()` [6]
- b) Write a program to display a string and count the number of characters in the string using a loop. [5]
- c) Explain the `ord()` and `chr()` functions with examples. Explain `in` and `not in` operators on strings. [4]

Unit V --- Object Oriented Programming (18 Marks)

Q5) Attempt the following: [15]

- a) Explain the following **programming paradigms**: i) **Monolithic Programming** ii) **Procedural Programming** iii) **Object Oriented Programming** [6]
- b) Explain the following OOP concepts with examples: i) **Class variable** ii) **Object variable** [5]
- c) Write a Python program to create a class `Employee` with attributes `Name`, `emp_id`, and `salary`. Display data of 4 employees. [4]

OR Q6) Attempt the following: [15]

- a) Explain the following features with examples: i) **Containership** ii) **Inheritance** iii) **Data Abstraction** [6]
- b) Explain **class method** and **self object** with suitable example. [5]
- c) Write a program to create a class `Car` with two attributes `name` and `cost` . Create two objects and display their information. [4]
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Unit VI --- File Handling (17 Marks)

- Q7)** Attempt the following: [15]
- a) What is a **file**? Explain **relative** and **absolute path** of a file with examples. [6]
- b) Explain the following file handling methods: i) `read(n)` ii) `readlines()` iii) `close()` [5]
- c) Explain the following file **access modes**: `r` , `w+` , `a+` , `rb` , `ab+` . [4]
- OR Q8)** Attempt the following: [15]
- a) Explain any three **directory methods** with examples. [6]
- b) Explain **text** and **binary files** with examples. [5]
- c) Explain any three **dictionary methods** with examples. [4]
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*** END OF QUESTION PAPER ***